Here the customer can place multiple orders and the order is related to only one customer. See Figure 3.11



Figure: 3.11 Cardinality Relation n to n

The example of many-to-many relationship is Students registering the Courses. A student can register more than one courses and A course can be registered by many students. Hence it is many-to-many.

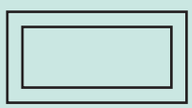
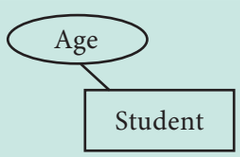
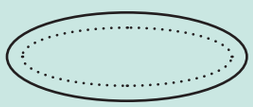
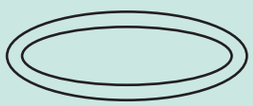
3.6 ER-Diagram

ER Diagram presents data visually on how they are related to each other. This model follows separate notations for representing data into entities, attributes and relationship between the entities.

3.6.1 ER-Modeling Diagram Notations

Entities, Attributes and Relationship forms the components of ER Diagram and the defined symbols and shapes are summarized below in Table 3.6.

Table 3.6 ER diagram Notations

ER Component	Description (how it is represented)	Notation
Entity - Strong	Simple rectangular box	
Entity – Weak	Double rectangular boxes	
Relationships	Rhombus symbol - Strong	
between Entities	Rhombus within rhombus – Weak	
Attributes	Ellipse Symbol connected to the entity	
Key Attribute for Entity	Underline the attribute name inside Ellipse	
Derived Attribute for Entity	Dotted ellipse inside main ellipse	
Multivalued Attribute for Entity	Double Ellipse	

XII CompApp_English version CHAPTER 03.indd 65 23-02-2020 11:09:47